
CLEAN FEATURE ANCHORING: MITIGATING THE ACCURACY-ROBUSTNESS TRADE-OFF BY SELF-DISTILLING A NATURALLY TRAINED MODEL

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ABSTRACT

Adversarial training improves robustness but often sacrifices accuracy on clean data, leading to a persistent clean-robust trade-off. We address this trade-off by first learning a high-accuracy clean self-teacher exclusively from natural images and then preserving its clean knowledge while adversarially training the student. We propose Clean Feature Anchoring (CFA), which matches the student’s adversarial representation to the fixed clean representation of the self-teacher while retaining its classifier, requiring no label-based objective during adversarial training.

We show that this formulation can be understood as direct regression onto a fixed clean target rather than predictive distillation through softened probabilities. For the feature-space realization, controlling the anchor jointly bounds clean teacher-student discrepancy and local feature variation. We further show that teacher clean accuracy alone does not determine how much clean performance is retained after robust transfer.

To account for example-dependent responses to perturbations, we additionally allocate a fixed batch budget of attack radii according to the local sensitivity of the anchoring objective. On CIFAR-100 with ResNet-18, sensitivity matching improves clean accuracy by 2.19 percentage points on average across three runs while leaving AutoAttack accuracy nearly unchanged. CFA achieves $64.82 \pm 0.07\%$ clean accuracy and $28.04 \pm 0.09\%$ AutoAttack accuracy, and the resulting clean-robust improvements extend to CIFAR-10, Tiny-ImageNet, and WideResNet-34-10.

1 INTRODUCTION

Adversarial training improves worst-case accuracy but often sacrifices clean accuracy (Madry et al., 2017; Tsipras et al., 2018; Zhang et al., 2019). Robust distillation can soften this trade-off, but requires an adversarially trained teacher (Goldblum et al., 2020; Zhu et al., 2021; Zi et al., 2021; Huang et al., 2023; Lee et al., 2025). A naturally trained teacher is cheaper and more accurate on clean inputs, yet vulnerable to attack. What should a robust student learn from such a teacher?

Many distillation methods express supervision as a predictive distribution and tune its softness or its weight relative to labels (Goldblum et al., 2020; Zi et al., 2021; Wu et al., 2024; Zhao et al., 2024). Direct regression offers a different view: fix what the teacher produces at the clean input and enforce that target throughout the student’s adversarial neighborhood. We study this principle with the teacher’s clean feature as the target, an inherited fixed classifier, and no label loss. This design exposes two further choices: which natural checkpoint supplies the target, and where to allocate the perturbation budget.

Clean Feature Anchoring (CFA) makes the target explicit: the student backbone matches one clean teacher representation even at adversarial inputs. Softening the teacher’s predictions mainly moves robustness, whereas direct clean-logit and feature regression improve both axes without a temperature search. Their near-tie identifies fixed clean-target regression, rather than the choice

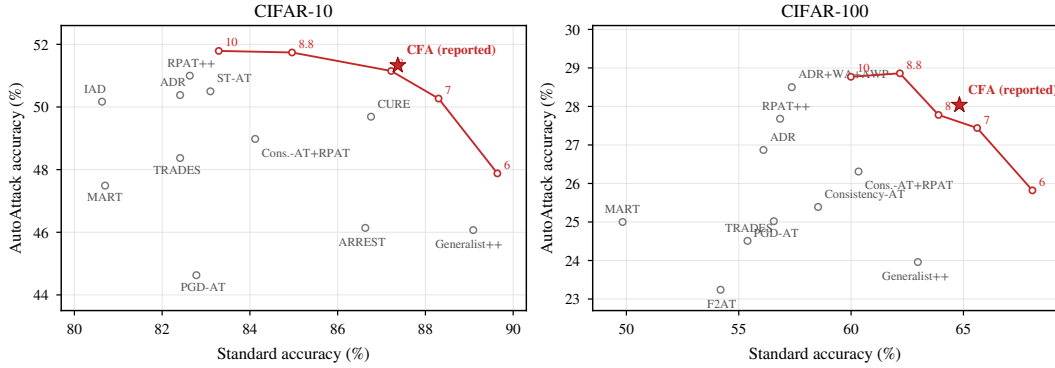


Figure 1: Clean–AutoAttack trade-off on CIFAR-10 and CIFAR-100 with ResNet-18 at $\epsilon = 8/255$. Grey points are published results (Table 29); the CFA curve varies the training radius over 100-epoch runs, and the star denotes the reported 150-epoch result trained at $\epsilon_{\text{train}} = 8/255$ (the three-run mean on CIFAR-100).

between logit and feature coordinates, as the broader mechanism. We use features because they retain the representation supplied to the inherited classifier. A fixed-anchor bound then relates the CFA loss to clean teacher fidelity and local feature variation without requiring the teacher to be robust.

The teacher checkpoint changes what the student learns. Across two independent CIFAR-100 training trajectories, later natural checkpoints shift students toward higher robustness and lower clean accuracy. Moreover, label-smoothed and mixup teachers with nearly identical clean accuracy produce students 4.18 clean points apart. Teacher accuracy alone is therefore not a sufficient description of the supervision a checkpoint provides.

Even with the teacher fixed, a uniform pixel radius need not perturb the anchor loss equally across examples. We allocate a fixed batch budget inversely to the loss’s input sensitivity. Across three runs of the final recipe, this changes clean accuracy from 62.63% to 64.82% while AA remains at 28.0%. A separate control shows that reassigning the same radii by sample difficulty weakens the result. The teacher determines the reference, while radius allocation determines where to enforce it.

CFA improves the measured trade-off across CIFAR-10, CIFAR-100, and Tiny-ImageNet. On CIFAR-100 with ResNet-18 it reaches 64.82% clean and 28.04% AA on average, the highest AutoAttack accuracy and NRR among twelve baselines run in the same framework (Table 3). Target, recipe, and radius controls distinguish the anchor’s contribution from that of the training stack.

Our contributions are:

- A controlled comparison from softened predictive targets to direct clean-target regression, instantiated as label-free CFA; for this feature realization, one anchor jointly bounds clean teacher fidelity and local variation.
- Evidence that the teacher checkpoint sets the student’s operating point rather than its quality: two independent trajectories move clean accuracy and robustness in opposite directions, and two teachers of nearly equal clean accuracy produce students 4.18 clean points apart.
- A radius allocation read from the loss’s own input sensitivity, which holds at the reported configuration and survives controls that keep the budget, and even the multiset of radii, fixed.

2 RELATED WORK

Adversarial training and its targets. PGD adversarial training optimizes a model on worst-case inputs (Madry et al., 2017); TRADES separates clean prediction from local consistency (Zhang et al., 2019), and MART reweights misclassified examples (Wang et al., 2020). Subsequent work

improves generalization through label smoothing, consistency regularization, weight averaging, and adversarial weight perturbation (Pang et al., 2020; Tack et al., 2022; Rice et al., 2020; Wu et al., 2020). ADR replaces one-hot labels with targets produced by an annealed momentum teacher (Wu et al., 2024). These approaches motivate our study of how the supervisory target affects the clean-robust trade-off.

Predictive targets in adversarial distillation. ARD transfers a robust teacher’s predictions during adversarial training (Goldblum et al., 2020). RSLAD emphasizes robust soft labels (Zi et al., 2021), IAD discounts unreliable teacher guidance (Zhu et al., 2021), and AdaAD uses the teacher in both adversarial-example generation and student optimization (Huang et al., 2023). IGDM adds the teacher’s local input geometry to logit-based distillation (Lee et al., 2025). These methods are designed to inherit robustness from an adversarially trained teacher. Our target instead comes from a natural teacher at the clean input and remains fixed during the student’s inner maximization.

Guidance from naturally trained models. BGAT is the closest direct-regression precedent: it matches adversarial logits to a fixed natural teacher’s clean logits, and LBGAT extends this to a jointly trained natural branch (Cui et al., 2021). Our logit control confirms that this fixed-clean-target principle is not specific to features. CFA uses the full clean representation with an inherited fixed classifier, and our analysis focuses on what a single anchor controls, which teacher checkpoint supplies it, and how its neighborhood is allocated. ARREST starts from a naturally pretrained model and regularizes latent representations while retaining adversarial cross-entropy and noisy replay (Suzuki et al., 2023). HAT uses a natural helper to label examples outside the training ball (Rade & Moosavi-Dezfooli, 2022), and B-MTARD combines clean and robust teachers with adaptive temperature and loss balancing (Zhao et al., 2024).

Sample-dependent perturbation budgets. IAAT and CAT adapt the training radius to sample difficulty, while MMA directly increases an input-space margin (Balaji et al., 2019; Cheng et al., 2022; Ding et al., 2020). Our allocation has a different source: it redistributes a fixed batch budget according to the input sensitivity of the loss being optimized. The comparison is therefore not between adaptive and uniform radii alone, but between the signals used to assign the same budget.

3 ANALYSIS

Adversarial training often improves robustness at the cost of clean accuracy. To mitigate this trade-off, we explore a self-distillation framework that first learns a clean self-teacher exclusively from natural images and then transfers its knowledge to a student during adversarial training. The clean self-teacher provides a high-accuracy representation learned without adversarial perturbations, while the student is optimized for robustness. A central question is therefore how this clean knowledge should be transferred under adversarial perturbations.

We first revisit predictive distillation, where the teacher’s softened output distribution serves as the supervision target, as commonly adopted in adversarial distillation. We then move from predictive supervision to a fixed clean target and study the feature-space realization used by CFA. Our analysis characterizes what this fixed anchor controls and subsequently examines whether improved clean performance of the self-teacher translates into improved performance of the adversarially trained student.

3.1 PRELIMINARIES

Let $x \in \mathcal{X} = [0, 1]^D$ denote a clean input, and define its ℓ_∞ -bounded perturbation set as

$$B(x, \epsilon) = \{x' \in \mathcal{X} : \|x' - x\|_\infty \leq \epsilon\}, \quad (1)$$

where x' denotes a perturbed input within the ϵ -neighborhood of x . During adversarial training, x' is chosen to maximize the corresponding student loss over $B(x, \epsilon)$.

We decompose a network as $f = h \circ \Phi$, where $\Phi : \mathcal{X} \rightarrow \mathbb{R}^d$ is a continuous feature map and $h : \mathbb{R}^d \rightarrow \mathbb{R}^C$, $h(z) = Wz + b$, is the classifier for C classes. We use subscripts t and s for the frozen clean self-teacher and the student, respectively. Both networks share the same architecture, with fixed network statistics for the analysis; feature distances are Euclidean unless stated otherwise.

3.2 FROM SOFT PREDICTIVE TARGETS TO A FIXED CLEAN TARGET

Soft-label robust distillation transfers the clean teacher’s knowledge through its predictive distribution, typically controlled by a temperature parameter τ (Goldblum et al., 2020; Zi et al., 2021; Wu et al., 2024; Zhao et al., 2024). Given a clean input x and a perturbed input $x' \in B(x, \epsilon)$, we define

$$q_t^\tau(x) = \text{softmax}(f_t(x)/\tau), \quad p_s^\tau(x') = \text{softmax}(f_s(x')/\tau), \quad (2)$$

where the teacher is evaluated on the clean input x , whereas the student is evaluated on its perturbed counterpart x' .

Conventional knowledge distillation applies the same temperature to both distributions and minimizes the scaled divergence

$$\mathcal{L}_{\text{KD}}^\tau(x, x') = \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')). \quad (3)$$

Increasing τ makes both predictive distributions progressively softer. In particular, the clean teacher target $q_t^\tau(x)$ approaches the uniform distribution as $\tau \rightarrow \infty$, and its entropy approaches the maximum value $\log C$, as illustrated in Figure 2a. This does not imply that the shared-temperature KD objective loses all information about the teacher. Although both distributions become nearly uniform, their deviations from uniformity still encode the relative structure of the underlying logits.

More precisely, for fixed logits and

$$\bar{f} = f - C^{-1} \mathbf{1} \mathbf{1}^\top f,$$

the conventional τ^2 scaling preserves this residual information in the high-temperature limit:

$$\lim_{\tau \rightarrow \infty} \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')) = \frac{1}{2C} \|\bar{f}_t(x) - \bar{f}_s(x')\|_2^2. \quad (4)$$

Thus, while the softened probabilities themselves approach uniformity, their scaled discrepancy converges to the squared distance between the centered teacher and student logits. The centering reflects the shift invariance of softmax: adding the same constant to every logit does not change the predictive distribution. Consequently, high-temperature predictive distillation approaches direct regression of the perturbed student’s logits onto the clean self-teacher’s logits, up to a common shift.

This connection motivates a more direct view of knowledge transfer. Rather than passing the clean teacher’s knowledge through a softened predictive distribution, the student can be directly regressed onto a fixed clean target. The high-temperature limit provides a logit-level realization of this principle. CFA instead applies the fixed-target principle at the representation level by anchoring the student’s perturbed feature to the clean self-teacher feature:

$$\mathcal{L}_{\text{anchor}}(x) = \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2. \quad (5)$$

Here, $\Phi_t(x)$ is computed from the clean input and serves as the fixed target, whereas the student is optimized against the worst-case perturbed input within $B(x, \epsilon)$. Logit regression constrains the representation through the classifier output, whereas feature regression directly constrains the representation supplied to the inherited classifier.

We next compare these forms of supervision to examine how the transition from predictive distillation to direct clean-target regression affects the clean–robust trade-off. All variants are trained for 50 epochs under an ℓ_∞ perturbation budget of $\epsilon = 8/255$.

As shown in Figure 2b, increasing the shared temperature moves the KD solution toward the operating point obtained by direct logit regression, consistent with the high-temperature limit in Equation (4). This occurs even as the teacher target becomes nearly uniform at large τ (Figure 2a), because the scaled discrepancy retains the relative structure of the underlying logits.

Direct feature regression yields 0.13 and 0.19 percentage-point improvements in clean accuracy and AA, respectively, over logit regression. Under the final training recipe, the corresponding differences remain small at 0.26/0.13 points (Table 6). Thus, direct regression onto a fixed clean target provides a temperature-free alternative to predictive distillation, while the small gap between logit and feature regression does not by itself establish an inherent empirical advantage of feature coordinates.

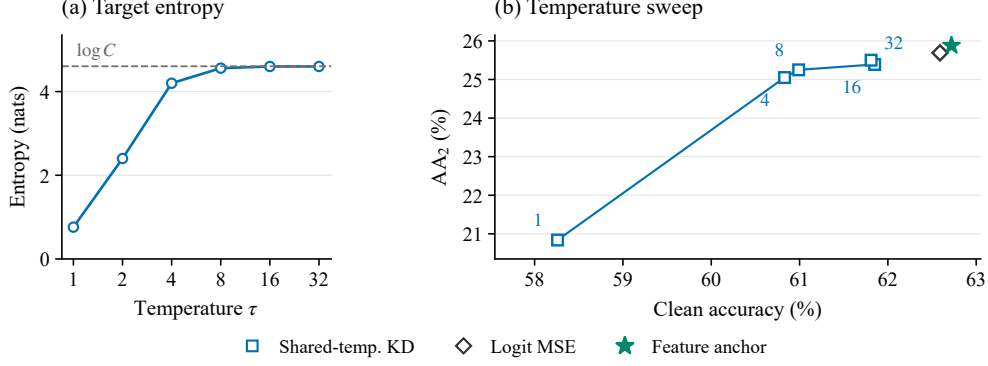


Figure 2: Target comparison on CIFAR-100 (ResNet-18). (a) Entropy of the clean teacher target $q_t^r(x)$, which approaches the uniform entropy $\log C$ as the shared temperature τ increases. (b) Clean accuracy and AA under shared-temperature KD for different values of τ , together with direct logit and feature regression. Numbers denote τ . The full temperature sweep is reported in Table 11.

We also note that KD updates the classifier, whereas the regression variants keep it fixed. The comparison therefore evaluates complete training designs rather than isolating the target representation alone.

The comparison shifts our focus from how a predictive distribution should be softened to what is controlled by anchoring the student to a fixed clean target. We next characterize this fixed-anchor property for the feature-space realization used by CFA.

3.3 WHAT THE ANCHOR CONTROLS

CFA is designed to preserve the representation of the clean self-teacher while the student is trained against adversarial perturbations. We therefore ask what controlling the fixed-anchor objective implies about the student’s representation. In particular, a desirable student should remain faithful to the clean teacher at the original input while exhibiting limited feature variation within the perturbation set.

To formalize these properties, let $x \in \mathcal{X}$ and $B = B(x, \epsilon)$, and define

$$L = \max_{x' \in B} \|\Phi_s(x') - \Phi_t(x)\|_2, \quad F = \|\Phi_s(x) - \Phi_t(x)\|_2, \quad O = \max_{x', x'' \in B} \|\Phi_s(x') - \Phi_s(x'')\|_2. \quad (6)$$

Here, L is the worst-case discrepancy from the fixed clean teacher anchor and corresponds to the quantity directly controlled by the CFA objective. F measures the clean feature discrepancy between the student and its self-teacher, capturing the fidelity of the transferred clean representation. O measures the diameter of the student’s feature set over the perturbation region and thus its local feature variation.

These quantities satisfy the following relation.

Proposition 1 (Fixed-anchor decomposition). *For every $x \in \mathcal{X}$ and $\epsilon \geq 0$,*

$$L \leq F + O \leq 3L. \quad (7)$$

Proof. For any $x' \in B$, the triangle inequality gives

$$\|\Phi_s(x') - \Phi_t(x)\|_2 \leq \|\Phi_s(x') - \Phi_s(x)\|_2 + \|\Phi_s(x) - \Phi_t(x)\|_2 \leq O + F.$$

Maximizing over $x' \in B$ yields $L \leq F + O$.

Conversely, since $x \in B$,

$$F = \|\Phi_s(x) - \Phi_t(x)\|_2 \leq L.$$

Moreover, for any $x', x'' \in B$,

$$\begin{aligned} \|\Phi_s(x') - \Phi_s(x'')\|_2 &\leq \|\Phi_s(x') - \Phi_t(x)\|_2 + \|\Phi_s(x'') - \Phi_t(x)\|_2 \\ &\leq 2L. \end{aligned}$$

Hence $O \leq 2L$, and therefore $F + O \leq 3L$. $\square$

The proposition gives a direct interpretation of the CFA objective. If $L \leq \delta$, then

$$F \leq \delta, \quad O \leq 2\delta.$$

Thus, keeping the worst-case student feature close to the clean teacher anchor simultaneously keeps the student’s clean representation close to the teacher and limits feature variation throughout the perturbation region. Importantly, this conclusion does not require the clean teacher itself to be robust or locally stable; only its representation at the clean input $\Phi_t(x)$ serves as the anchor.

Because CFA inherits the teacher’s linear classifier, the same discrepancy also bounds the deviation at the classifier output. For $h_t(z) = W_t z + b_t$,

$$\|h_t(\Phi_s(x')) - h_t(\Phi_t(x))\|_2 \leq \|W_t\|_{2 \rightarrow 2} \|\Phi_s(x') - \Phi_t(x)\|_2 \leq \|W_t\|_{2 \rightarrow 2} L. \quad (8)$$

A small anchor discrepancy therefore limits how far the student’s logits can deviate from the clean teacher logits under the inherited classifier. This bound alone does not guarantee identical predictions, which additionally depend on the teacher’s clean classification margin.

The fixed-anchor argument is not specific to feature coordinates and analogously applies to norm-based regression onto a fixed logit target. It therefore characterizes the joint control induced by a fixed target rather than establishing an inherent advantage of feature matching. In practice, finite-step PGD only approximates the inner maximization and thus does not certify the true value of L .

3.4 DOES BETTER CLEAN TEACHER PERFORMANCE TRANSFER TO THE STUDENT?

The preceding analysis shows how a fixed clean self-teacher representation can serve as an anchor for the adversarially trained student. This raises a natural question about the source of that target: does improving the clean performance of the self-teacher also improve the clean performance transferred to the student?

A straightforward expectation is that a higher-accuracy clean teacher provides a better clean target and therefore yields a student with higher clean accuracy. We first examine this relationship using different checkpoints along the training trajectory of the same clean self-teacher. Each checkpoint provides the student initialization, feature target, and inherited classifier, allowing us to observe how changes in teacher clean performance are reflected in the resulting student.

As the clean self-teacher is trained from 50 to 300 epochs, its clean accuracy increases by 2.51 percentage points. This improvement does not translate into higher student clean accuracy. Instead, student clean accuracy decreases by 2.04 points, while AA increases from 24.38% to 25.80% (Figure 3a). Thus, improvements in the teacher’s clean accuracy are not directly inherited by the adversarially trained student; different teacher states instead lead to different clean–robust operating points. The relationship is also not strictly monotonic, as the student transferred from the 200-epoch teacher achieves slightly higher AA than that transferred from the 300-epoch teacher.

To examine whether this behavior is specific to a single teacher trajectory, we repeat the experiment with an independently trained clean self-teacher. The same overall pattern emerges. From 40 to 300 teacher epochs, teacher clean accuracy increases from 75.57% to 78.27%, whereas student clean accuracy decreases from 64.36% to 62.34% and AA increases from 23.86% to 25.94% (Table 1). Across two independent trajectories, higher teacher clean accuracy therefore does not directly translate into higher student clean accuracy after robust transfer.

Teacher clean accuracy alone does not reveal what changes in the transferred representation as training progresses. To examine the representation associated with this shift, we measure the within-between scatter ratio S_w/S_b on unit-normalized clean features. As shown in Figure 3b, the teacher and student ratios decrease together as teacher training progresses, indicating increasingly separated

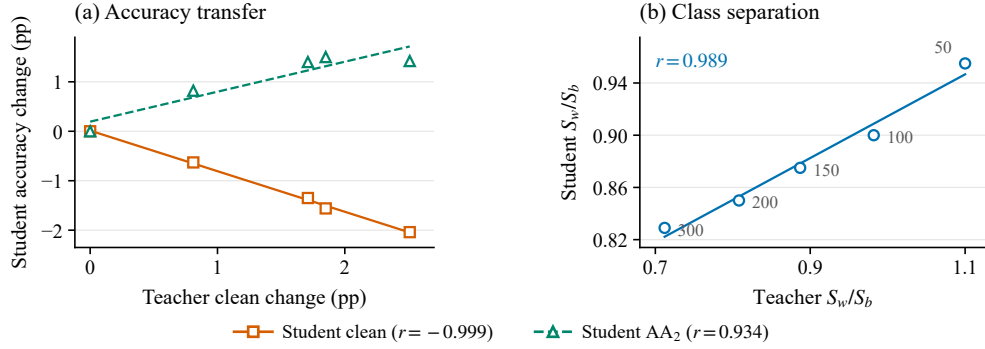


Figure 3: Transfer from clean self-teachers at different stages of training on CIFAR-100 (ResNet-18). (a) Changes in teacher clean accuracy and resulting student clean accuracy and AA relative to the 50-epoch teacher. (b) Within-to-between scatter ratios S_w/S_b of teacher and student features; lower values indicate stronger class separation, and labels denote teacher epochs. Lines are least-squares fits. Exact values are reported in Table 12.

Table 1: Transfer from an independently trained clean self-teacher trajectory on CIFAR-100 (ResNet-18; 50-epoch students). Full results are reported in Table 24.

Teacher epoch	Teacher clean	Student clean	Student AA	Student NRR
40	75.57	64.36	23.86	34.81
100	77.27	63.68	25.61	36.53
200	77.90	63.02	26.16	36.97
300	78.27	62.34	25.94	36.64

class structure in both representations. This association suggests that changes in the teacher representation are reflected in the resulting student, but does not identify which representational property causes the observed clean-robust shift.

The checkpoint sweep changes teacher accuracy together with training stage and the learned representation. We therefore further examine whether teacher clean accuracy alone is sufficient to characterize transfer. To this end, we compare clean self-teachers trained with label smoothing and mixup that achieve nearly identical clean accuracy.

Table 2: Transfer from accuracy-matched clean self-teachers on CIFAR-100 (ResNet-18; 50-epoch students). Feature sensitivity is measured on held-out data.

Natural training	Teacher		Student	
	Clean	Sensitivity	Clean	AA
Label smoothing 0.1	78.76	3.16	59.62	25.72
Mixup	78.62	4.82	63.80	25.70

Despite differing by only 0.14 percentage points in teacher clean accuracy, the two teachers produce students whose clean accuracies differ by 4.18 points, while their AA differs by only 0.02 points (Table 2). Together with the checkpoint trajectories, this result shows that teacher clean accuracy alone is insufficient to determine the clean performance retained by the adversarially trained student.

The teachers also differ in feature sensitivity in the same order as student clean accuracy, suggesting that properties of the transferred representation beyond predictive accuracy may be relevant. With only one accuracy-matched pair, however, this observation is insufficient to establish feature sensitivity as a general predictor of transfer.

Taken together, these results distinguish teacher clean performance from transferability. A more accurate clean self-teacher does not necessarily produce a student that retains more clean accuracy after adversarial training, and teachers with nearly identical clean accuracy can transfer substantially

different clean performance. Teacher clean accuracy is therefore an incomplete descriptor of the representation being transferred rather than a sufficient criterion for selecting a teacher.

For the remaining experiments, we fix the 200-epoch checkpoint as a reference teacher and hold it constant. We then examine a source of variation that persists even with the teacher fixed: the same perturbation radius need not induce the same anchor discrepancy across individual examples.

4 METHOD

The preceding analysis motivates two design choices for transferring clean knowledge during adversarial training. First, the clean self-teacher provides a fixed representation target, allowing the student to preserve the teacher’s clean representation while controlling its variation under perturbations. Second, even with the teacher fixed, a uniform input-space radius need not induce a comparable change in the anchoring objective across examples. Based on these observations, we construct CFA by combining clean feature anchoring with sample-dependent perturbation budgets. We first describe the feature-anchoring objective and then introduce a sensitivity-based rule for allocating the perturbation radius across training examples.

4.1 CLEAN FEATURE ANCHORING

We first instantiate the fixed-target principle by using the clean self-teacher’s feature representation as the anchor for adversarial training. Let Φ_t and h_t denote the frozen feature map and classifier of the clean self-teacher. The student inherits both, $\Phi_s \leftarrow \Phi_t$ and $h_s = h_t$, after which only Φ_s is optimized.

For a clean example x , CFA solves

$$\begin{aligned} x_{\text{adv}} &= \arg \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2, \\ \mathcal{L}_{\text{CFA}} &= \mathbb{E}_x [\|\Phi_s(x_{\text{adv}}) - \Phi_t(x)\|_2^2]. \end{aligned} \quad (9)$$

We approximate the inner maximization with randomly initialized PGD against the fixed clean target $\Phi_t(x)$ and minimize the same discrepancy over the student backbone. This adversarial-training stage uses no label loss, while inference uses $h_t \circ \Phi_s$. Algorithm 1 summarizes the complete procedure.

4.2 SENSITIVITY-MATCHED ϵ

The anchoring objective above still leaves the perturbation radius to be specified. A uniform pixel radius gives every example the same input-space budget, but does not account for how strongly the anchor loss responds to that perturbation. We therefore allocate the perturbation budget according to the local sensitivity of the anchoring objective. The mean radius within each batch is kept fixed, so the procedure redistributes rather than increases the overall training budget.

For

$$\ell_{\text{anc}}(x'; x_i) = \|\Phi_s(x') - \Phi_t(x_i)\|_2^2,$$

consider its first-order change around the clean input x_i . Under an ℓ_∞ perturbation of radius e ,

$$\max_{\|\delta\|_\infty \leq e} \left\langle \nabla_{x'} \ell_{\text{anc}}(x'; x_i) \Big|_{x'=x_i}, \delta \right\rangle = e \left\| \nabla_{x'} \ell_{\text{anc}}(x'; x_i) \Big|_{x'=x_i} \right\|_1. \quad (10)$$

Thus, the local increase in anchor loss scales with both the perturbation radius and the sensitivity of the anchoring objective.

Motivated by this relation, let $g_i > 0$ denote a local sensitivity score for example x_i . We assign smaller perturbation radii to more sensitive examples and larger radii to less sensitive ones according to

$$\epsilon_i = N \epsilon \frac{g_i^{-p}}{\sum_{j=1}^N g_j^{-p}}, \quad (11)$$

where N is the batch size and ϵ is the mean perturbation radius. The exponent p controls the strength of sensitivity matching: $p = 0$ recovers a uniform radius, while larger p increasingly allocates smaller radii to more sensitive examples.

By construction,

$$\sum_{i=1}^N \epsilon_i = N\epsilon,$$

so sensitivity matching changes only how the perturbation budget is distributed across examples, rather than the average strength of adversarial training. Unlike sample weighting in the outer objective, this allocation changes the perturbation neighborhood assigned to each example while retaining equal weight for each example in the training average.

The first-order relation in Equation (10) motivates sensitivity as the quantity governing radius allocation, rather than asserting that the resulting nonlinear adversarial losses are exactly equalized. The key empirical question is therefore whether sensitivity provides a useful assignment of the perturbation budget. Accordingly, Table 7 compares sensitivity matching with a uniform radius and alternative assignment rules, including a control that preserves the same multiset of sensitivity-derived radius multipliers but reassigns them according to sample difficulty. If the gain arises merely from introducing heterogeneous radii, this reassignment should preserve the improvement; otherwise, which example receives each radius matters.

5 EXPERIMENTS

We evaluate CFA with three questions in mind: how it compares with existing adversarial training and robust distillation methods, whether its performance generalizes across datasets and model architectures, and how its individual design components contribute to the resulting clean-robust trade-off.

5.1 EXPERIMENTAL SETTINGS

We evaluate CFA on CIFAR-10, CIFAR-100 (Krizhevsky et al., 2009), and Tiny-ImageNet-200 (Le & Yang, 2015), using ResNet-18 as the default architecture. We additionally evaluate WideResNet-34-10 to examine generalization across model capacity.

Unless stated otherwise, the clean self-teacher is trained on natural images for 200 epochs and then frozen. The student is initialized from the teacher, with the inherited classifier kept fixed during adversarial training. Following Section 4, the student is trained using clean feature anchoring with sensitivity-matched perturbation radii ($p = 1$). The main CIFAR experiments use 150 training epochs and $\epsilon_{\text{train}} = 8/255$. We use AdamW with a one-cycle learning-rate schedule and 10-step PGD, together with weight averaging (WA) and adversarial weight perturbation (AWP). Teacher statistics and dataset- and architecture-specific training settings are reported in Table 22 and appendix F.

We compare CFA with standard adversarial training and robust distillation baselines using their respective training recipes. Distillation-based baselines receive the same clean self-teacher where applicable. We report clean accuracy and AutoAttack (AA) accuracy on the full test set at $\epsilon = 8/255$, together with their arithmetic mean (Avg) and harmonic mean (NRR). Additional evaluation details, attack diagnostics, and results with alternative teachers are provided in Appendices F and I and table 29.

5.2 MAIN RESULTS

Table 3 compares CFA with standard adversarial training and robust distillation methods on CIFAR-10 and CIFAR-100. On CIFAR-100, CFA achieves 64.82% clean accuracy and 28.04% AA, yielding an NRR of 39.15. Compared with ADR, the closest baseline in NRR, CFA improves clean accuracy by 4.89 points, AA by 0.84 points, and NRR by 1.73 points. ARREST achieves higher clean accuracy at 66.97%, but its AA is 5.26 points lower than CFA, reflecting a different clean-robust operating point.

On CIFAR-10, CFA reaches 87.37% clean accuracy and 51.33% AA, with an NRR of 64.67. Its NRR exceeds the closest baseline by 1.85 points. Together, these results show that CFA maintains high clean accuracy while remaining competitive in robust accuracy across both CIFAR datasets.

Table 3: ResNet-18 results under ℓ_∞ perturbations with $\epsilon = 8/255$. All values are measured in our framework. T denotes the teacher assumed by each method: none (–), a natural model (nat.), or the method’s own weight average (self). The CIFAR-100 CFA row reports the mean of three student seeds; all other rows are individual runs.

Method	T.	CIFAR-10				CIFAR-100			
		Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	–	84.54	43.57	64.06	57.50	57.36	20.34	38.85	30.03
TRADES	–	81.71	49.16	65.44	61.39	55.28	23.55	39.42	33.03
MART	–	80.41	45.56	62.99	58.16	53.86	22.72	38.29	31.96
ARD	nat.	84.58	43.75	64.16	57.67	57.61	20.24	38.93	29.96
RSLAD	nat.	85.15	42.91	64.03	57.06	59.68	21.30	40.49	31.40
AdaAD	nat.	85.39	46.22	65.81	59.98	59.79	23.19	41.49	33.42
AdaAD + IGDM	nat.	85.01	45.50	65.26	59.27	48.25	19.48	33.87	27.75
LBGAT	nat.	–	–	–	–	56.46	26.02	41.24	35.62
ARREST	nat.	85.06	46.42	65.74	60.06	66.97	22.78	44.88	34.00
HAT	nat.	84.59	48.81	66.70	61.90	60.67	24.45	42.56	34.85
ADR	self	83.36	49.72	66.54	62.29	59.93	27.20	43.57	37.42
RPAT++	–	82.41	50.75	66.58	62.82	55.93	27.36	41.64	36.74
CFA (ours)	nat.	87.37	51.33	69.35	64.67	64.82	28.04	46.43	39.15

5.3 GENERALIZATION ACROSS DATASETS AND ARCHITECTURES

We next examine whether the behavior observed on the CIFAR benchmarks extends to a larger-scale dataset and a higher-capacity architecture.

Table 4: Results on Tiny-ImageNet-200 with ResNet-18 and CIFAR-10 with WideResNet-34-10, measured in our framework.

Method	T.	Tiny-ImageNet, ResNet-18				CIFAR-10, WRN-34-10			
		Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	–	49.72	14.45	32.09	22.39	86.00	45.71	65.86	59.69
TRADES	–	46.98	15.95	31.46	23.83	85.13	48.99	67.06	62.19
MART	–	46.01	16.90	31.46	24.73	85.29	47.06	66.18	60.65
ARD	nat.	49.74	14.48	32.11	22.43	86.35	45.52	65.94	59.61
RSLAD	nat.	51.26	17.02	34.14	25.56	86.41	45.80	66.10	59.87
AdaAD	nat.	51.59	16.79	34.19	25.33	86.80	46.84	66.82	60.85
LBGAT	nat.	51.04	18.47	34.75	27.13	81.56	51.79	66.68	63.35
ARREST	nat.	59.38	16.06	37.72	25.28	87.54	48.83	68.19	62.69
HAT	nat.	52.57	17.28	34.93	26.01	87.15	53.21	70.18	66.07
ADR	self	47.66	20.85	34.25	29.01	88.64	52.34	70.49	65.82
CFA (ours)	nat.	55.16	20.54	37.85	29.93	88.67	55.29	71.98	68.11

On Tiny-ImageNet-200, CFA achieves the highest Avg and NRR without maximizing either end-point individually (Table 4). ARREST attains higher clean accuracy, while ADR exceeds CFA in AA by 0.31 points. CFA instead reaches 55.16% clean accuracy and 20.54% AA, resulting in the highest NRR of 29.93.

The effect of the clean self-teacher observed in Section 3.4 is also present on Tiny-ImageNet. Replacing the 200-epoch reference teacher with an 80-epoch teacher changes the resulting student from 55.16/20.54 to 57.08/18.96 clean/AA. This again illustrates that changing the clean teacher can shift the student’s clean–robust operating point rather than uniformly improving both metrics.

With WideResNet-34-10 on CIFAR-10, CFA achieves 88.67% clean accuracy and 55.29% AA. Its clean accuracy is nearly identical to ADR (88.64%), while its AA is 2.95 points higher. On CIFAR-100 with the same architecture, CFA reaches 65.04/30.14 clean/AA, compared with 64.50/29.04 for ADR (Table 21). These results indicate that the observed behavior is not restricted to ResNet-18 or to a single dataset.

5.4 ABLATION AND COMPONENT ANALYSIS

We next examine the components underlying CFA. We first isolate the clean feature-anchoring objective, then compare feature and logit targets, and finally study the contribution and assignment of sample-dependent perturbation radii. We separately report the effects of weight averaging and AWP.

Clean feature anchoring. We first test whether the gain arises from the clean feature anchor rather than from the surrounding adversarial-training recipe. Under the same base training regime, replacing feature anchoring with label-based cross-entropy reduces both clean and robust accuracy.

Table 5: Objective control on CIFAR-100 with ResNet-18, 100 epochs, no weight averaging or AWP, $p = 0$, and $\epsilon_{\text{train}} = 8.8/255$.

Objective	Clean	AA	NRR
Anchor	61.29	25.36	35.88
Label cross-entropy, same recipe	54.60	19.45	28.68
Label cross-entropy, own recipe	56.66	20.91	30.55

As shown in Table 5, replacing the anchor objective with cross-entropy under the same recipe reduces clean accuracy by 6.69 points and AA by 5.91 points. Allowing cross-entropy to use its own optimizer and schedule narrows the gap, but CFA remains ahead by 4.63 clean and 4.45 AA points. Initialization and schedule controls change NRR by at most 0.22 (Table 25), indicating that the difference cannot be attributed solely to these training choices.

Feature and logit targets. The analysis in Section 3.2 showed that fixed-target regression need not be restricted to feature coordinates. We therefore compare feature anchoring with direct logit regression under otherwise comparable training conditions.

At a uniform radius of $8.8/255$ and 100 training epochs, logit regression reaches $60.36/28.02$ clean/AA, compared with $60.42/28.42$ for feature regression (Table 13). Under the final 150-epoch recipe with sensitivity-matched radii, logit regression reaches $64.49/27.85$ clean/AA (NRR 38.90), compared with $64.82/28.04$ (NRR 39.15) for feature regression (Table 6). The small differences are consistent with our analysis: effective transfer through a fixed clean target is not confined to feature space.

Sensitivity-matched radius allocation. We next isolate the second component of CFA: allocating the perturbation radius according to local sensitivity. We hold the teacher, training schedule, mean radius, weight averaging, and AWP fixed and vary only the radius allocation.

Table 6: Clean-target regression in the final CIFAR-100 recipe (ResNet-18, 150 epochs, $\epsilon_{\text{train}} = 8/255$). Feature rows report mean $\pm$ sample standard deviation over three student seeds and differ only in p ; logit regression is one run at $p = 1$.

Target and radius	Clean	AA	NRR
Features, uniform ($p = 0$)	62.63 ± 0.22	28.01 ± 0.20	38.71 ± 0.21
Features, sensitivity-matched ($p = 1$)	64.82 ± 0.07	28.04 ± 0.09	39.15 ± 0.09
Logits, sensitivity-matched ($p = 1$)	64.49	27.85	38.90

Replacing the uniform radius with sensitivity matching increases clean accuracy by 2.19 ± 0.20 points across three paired runs, while AA changes by only 0.03 ± 0.23 points (Table 6). The clean-accuracy improvement appears in every seed, whereas the AA difference is comparable to run-to-run variation. Thus, in the final training configuration, sensitivity matching primarily shifts the operating point toward higher clean accuracy without an observed loss in AA. This empirical behavior is not implied by the first-order motivation in Section 4.2, which only motivates how the perturbation budget is allocated.

Which signal should allocate the radius? The previous comparison establishes the effect of sensitivity matching relative to a uniform radius, but does not determine whether the sensitivity signal

itself matters or whether heterogeneous radii are sufficient. We therefore compare sensitivity with alternative assignments under the same mean perturbation budget.

Table 7: Radius-allocation signals under the same mean radius on CIFAR-100 with ResNet-18, trained for 100 epochs with WA and AWP. $\text{sd}(w)$ measures multiplier dispersion.

Allocation	$\text{sd}(w)$	Clean	AA	NRR
Uniform	0	60.42	28.42	38.66
Sensitivity	0.32	62.17	28.86	39.42
Same multipliers, difficulty order	0.32	60.90	28.06	38.42
Difficulty magnitude	0.39	61.27	28.12	38.55
Logit margin	0.42	61.07	27.79	38.20

Sensitivity matching gives the highest clean accuracy, AA, and NRR among the tested allocation signals (Table 7). In particular, reassigning the same set of sensitivity-derived radius multipliers according to sample difficulty reduces AA by 0.80 points and NRR from 39.42 to 38.42. Because this control preserves the radius distribution while changing which example receives each radius, the result indicates that the gain cannot be explained solely by introducing heterogeneous perturbation radii. Difficulty magnitude and logit-margin allocation also yield lower NRR than the sensitivity-based assignment.

Weight averaging and AWP. Finally, we examine the additional training components used in the final configuration. Weight averaging and AWP are not defining components of the CFA objective, but are included in the final recipe to strengthen adversarial training. In the ordered ablation, adding weight averaging and AWP increases AA by 1.95 and 1.17 points, respectively (Table 26). Because this ordered ablation does not isolate interactions between the two components, additional radius and training-stack comparisons are reported in Tables 17 and 18.

To verify that the final training stack alone does not account for CFA’s performance, we also apply teacher initialization, weight averaging, and AWP to the baseline methods.

Table 8: Adding teacher initialization, weight averaging, and AWP to baseline recipes on CIFAR-100 (ResNet-18).

Method	Own recipe			+ teacher init + WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
PGD-AT	57.36	20.34	30.03	59.85	25.77	36.03
TRADES	55.28	23.55	33.03	56.16	24.56	34.17
MART	53.86	22.72	31.96	50.97	25.13	33.66
ARD	57.61	20.24	29.96	60.19	26.10	36.41
RSLAD	59.68	21.30	31.40	63.44	25.55	36.43
AdaAD	59.79	23.19	33.42	57.21	27.10	36.78
AdaAD + IGDM	48.25	19.48	27.75	54.24	25.56	34.75
Consistency-AT	57.53	20.01	29.69	60.74	25.38	35.80
LBGAT	56.46	26.02	35.62	53.91	25.16	34.31
ADR	59.93	27.20	37.42	59.84	27.84	38.00
HAT	60.67	24.45	34.85	57.63	24.74	34.62
ARREST	66.44	22.99	34.16	64.95	23.02	33.99
CFA (ours)	—			64.82	28.04	39.15

The additional training components increase NRR for nine of the twelve baselines, but none reaches CFA’s NRR of 39.15 (Table 8); ADR is the closest at 38.00. This comparison indicates that the performance of CFA cannot be attributed solely to teacher initialization, weight averaging, or AWP. ARREST uses its 20-epoch recipe in this control, whereas Table 3 reports its 150-epoch comparison; ADR already includes a SAM-style weight perturbation in its original recipe.

Attack diagnostics, repeat-seed results, and additional implementation controls are provided in Appendices F and I.

6 CONCLUSION

A natural teacher can guide adversarial training through one clean target without a label loss. Its checkpoint changes the student’s clean–robust operating point, while sensitivity-based radii improve clean accuracy at nearly unchanged AutoAttack accuracy. Clean-logit regression reaches a similar endpoint, locating the broader mechanism in fixed clean-target regression rather than a uniquely superior feature metric. CFA thus separates the target, teacher, and perturbation budget.

A ALGORITHM

Algorithm 1 Clean Feature Anchoring (CFA)

```

1: Input: dataset  $\mathcal{D}$ , radius  $\epsilon$ , attack steps  $K$ , step size  $a$ , exponent  $p$ , clipping thresholds  $[w_{lo}, w_{hi}]$ 
2: Output: robust student  $h_t \circ \Phi_s$ 
3:
4: {Phase 1 — natural self-teacher. The only stage that uses labels.}
5: Train  $f_t = h_t \circ \Phi_t$  on  $\mathcal{D}$  with standard cross-entropy; freeze it
6:
7: {Phase 2 — anchored adversarial training. No labels, no head update.}
8:  $\Phi_s \leftarrow \Phi_t$  {student initialized at the teacher}
9: for epoch = 1 to  $T$  do
10:   for minibatch  $\{x_i\}_{i=1}^N \subset \mathcal{D}$  do
11:      $\phi_i \leftarrow \Phi_t(x_i)$  {teacher read once, at the clean input}
12:      $g_i \leftarrow \max(\|\nabla_x \|\Phi_s(x_i) - \phi_i\|_2, 10^{-12})$  {one backward pass}
13:      $v_i \leftarrow \text{clip}((\bar{g}/g_i)^p, w_{lo}, w_{hi})$ ;  $w_i \leftarrow N v_i / \sum_j v_j$ 
14:      $\epsilon_i \leftarrow \epsilon w_i$ ;  $a_i \leftarrow a w_i$  { $\sum_i \epsilon_i = N\epsilon$ }
15:      $x_i^0 \leftarrow x_i + \eta_i$ 
16:     for  $k = 0$  to  $K - 1$  do
17:        $x_i^{k+1} \leftarrow \Pi_{B(x_i, \epsilon_i)}(x_i^k + a_i \text{sign } \nabla_x \|\Phi_s(x_i^k) - \phi_i\|_2)$ 
18:     end for
19:      $\mathcal{L} \leftarrow \frac{1}{N} \sum_i \|\Phi_s(x_i^K) - \phi_i\|^2$ 
20:      $\Phi_s \leftarrow \Phi_s - \mu \nabla_{\Phi_s} \mathcal{L}$  { $h_t$  is never updated}
21:   end for
22: end for

```

B TWO PARTIAL RESULTS ON THE ROBUSTNESS SIDE

Natural-teacher guidance is shared with prior work; our question is whether a fixed clean target can support adversarial training without an additional label objective. In CFA, the teacher specifies the target while the student’s inner maximization enforces matching under perturbations. Teacher vulnerability alone therefore does not determine the student’s response: matching a clean target is different from reproducing the teacher at perturbed inputs.

The two results below examine this distinction in restricted settings. The first studies robust and non-robust features in a two-feature model; the second relates anchor error to a classification margin. They supplement the fixed-anchor decomposition, but do not predict the measured accuracy or establish superiority over other regression targets.

B.1 IN THE TWO-FEATURE MODEL, THE NON-ROBUST COEFFICIENT IS ZEROED RATHER THAN SHRUNK

The setting is the two-feature model of Tsipras et al. (2018), which is where the objection is usually posed. One feature x_1 agrees with the label with probability $p = 0.95$ and cannot be attacked. A further d features $z_j \sim \mathcal{N}(\eta y, 1)$ are individually almost uninformative, are jointly close to perfectly informative through their mean $m(z)$, and can all be moved by an ℓ_∞ adversary. A naturally trained network therefore reads $m(z)$: near-perfect clean accuracy, no robustness whatever. That is our teacher’s profile, so $\Phi_t = m(z)$.

Write the student’s feature as $\Phi_s = ax_1 + bm(z)$. Then b is exactly the quantity the objection is about: how much of the teacher’s fragile route the student takes over. If the objection is right, minimizing our objective should leave b large.

Proposition 2. *With the inner maximum exact, $J(a, b) = \mathbb{E}[R^2] + 2\epsilon|b| \mathbb{E}|R| + \epsilon^2 b^2$ with $R = ax_1 + (b - 1)m$. Its one-sided slopes at $b = 0$ differ by $4\epsilon \mathbb{E}|R(a, 0)|$; above a threshold ϵ_0 , every minimizer has $b^* = 0$.*

Proof. The adversary reaches Φ_s only through m , and only through the mean perturbation $\bar{\delta} = \frac{1}{d} \sum_j \delta_j$, which ranges over $[-\epsilon, \epsilon]$ as $\|\delta\|_\infty \leq \epsilon$. The inner objective is therefore $\max_{|\bar{\delta}| \leq \epsilon} (R + b\bar{\delta})^2 = (|R| + \epsilon|b|)^2$, since a scalar quadratic is maximized at an endpoint and the sign of $\bar{\delta}$ is free. Taking expectations and expanding the square gives the display.

Fix a and differentiate in b . The term $\mathbb{E}[R^2]$ and $\epsilon^2 b^2$ are smooth, while $2\epsilon|b| \mathbb{E}[R]$ contributes $\pm 2\epsilon \mathbb{E}[R]$ as $b \rightarrow 0^\pm$. The one-sided derivatives at $b = 0$ are thus $2\mathbb{E}[R(a, 0) m] \pm 2\epsilon \mathbb{E}[R(a, 0)]$, so $b = 0$ is a minimum along b exactly when

$$|\mathbb{E}[R(a, 0) m]| \leq \epsilon \mathbb{E}[R(a, 0)]. \quad (12)$$

At $b = 0$ the objective in a is $\mathbb{E}(ax_1 - m)^2$. Writing $q = 2p - 1$ for the mean label-alignment of x_1 , it is minimized at $a^* = \mathbb{E}[x_1 m] / \mathbb{E}[x_1^2] = q\eta$, and there $\mathbb{E}[R(a^*, 0) m] = q^2 \eta^2 - (\eta^2 + \frac{1}{d}) = -c$ with

$$c = \eta^2 v + \frac{1}{d}, \quad v = 1 - q^2. \quad (13)$$

Equation (12) therefore holds for every

$$\epsilon \geq \epsilon_0 = \frac{c}{\mathbb{E}[q\eta x_1 - m]}, \quad (14)$$

which is the stated threshold. For fixed x_1, m and $\bar{\delta} \in [-\epsilon, \epsilon]$, $(ax_1 + (b-1)m + b\bar{\delta})^2$ is convex in (a, b) . Its maximum over $\bar{\delta}$ and then its expectation are convex, so J is jointly convex. At $(a^*, 0)$ the a -derivative vanishes, and Equation (12) puts zero in the b -subgradient; hence this point is a global minimizer. If $\epsilon > \epsilon_0$, the directional derivative from $(a^*, 0)$ is strictly positive in every direction with nonzero b . Along $b = 0$, $\mathbb{E}(ax_1 - m)^2$ has a unique minimizer. Thus every global minimizer has $b = 0$ above the threshold. $\square$

Why the coefficient is deleted rather than shrunk. The adversary contributes $2\epsilon|b| \mathbb{E}[R]$ to the loss — linear in $|b|$, not quadratic in b . A quadratic penalty pushes a coefficient toward zero and never arrives; a penalty on $|b|$ is kinked at the origin and sets the coefficient to zero outright once its slope dominates, which is the same distinction that makes the lasso sparse and the ridge not. What produces the kink here is that the student is asked to reproduce a *value* rather than a *route*: the fragile features carry that value on clean data but not inside the ball, so past a certain radius carrying any of them costs more than it pays. The teacher supplies what to match, and the inner maximization discards how the teacher got there.

At $d = 512$ the model's constants give $c = 7.89 \times 10^{-3}$ and $\mathbb{E}[q\eta x_1 - m] = 0.0529$, hence $\epsilon_0 = 0.843 \eta$ — inside the bracket $(0.5\eta, 1.0\eta)$ that the Monte-Carlo minimization returns, and far below the 2η at which the model is usually stated.

The model assumes x_1 is unperturbable and z fully perturbable, so some suppression of z is built in. The proposition establishes the exact threshold behavior within this restricted model; it does not describe the geometry of real image classifiers.

That the exactness is the kink's doing, and not a general consequence of adding an adversary, is visible from dropping it. With the $|b|$ term removed, $J_{\text{quad}} = \mathbb{E}[R^2] + \epsilon^2 b^2$; minimizing out a and writing $u = 1 - b$ collapses it to $u^2 c + \epsilon^2(1 - u)^2$, so

$$b_{\text{quad}}^* = \frac{c}{c + \epsilon^2}, \quad (15)$$

which is strictly positive at every radius. Against the exact minimizer ($d = 512$, Monte-Carlo, $n = 2 \times 10^5$):

ϵ	b^* (exact)	b_{quad}^* (kink dropped)
0.5η	0.510	0.502
1.0η	0.000	0.202
2.0η	0.000	0.060
4.0η	0.000	0.016

The two columns agree below ϵ_0 and separate above it: the quadratic surrogate shrinks the non-robust coefficient and never deletes it, which is the ordinary picture, while the kinked objective sets it to zero and keeps it there.

Since every ϵ in J multiplies b , at $b^* = 0$ the objective reduces to $\mathbb{E}(ax_1 - m)^2$ and no longer depends on the radius. This saturation is a property of the two-feature model, not a prediction of the empirical radius sweep in Table 17.

The two-feature model cannot explain the measured clean-robust trade-off: it has only a robust and a fragile direction. It shows the narrower point that a robust student need not copy the fragile route by which its natural teacher computes the clean target.

B.2 WHAT ROBUST ACCURACY TRACKS IS THE LOSS MEASURED AGAINST A MARGIN

Writing $\gamma_t(x)$ for the teacher’s margin at x and D_y for the relevant decision-boundary constant, Cauchy-Schwarz gives that $\gamma_t(x) > D_y \cdot L(x)$ implies the student is robustly correct at x . The certificate is vacuous at the scales we train at, but the ratio DL/γ_t tracks both axes across the teacher ladder of Figure 3. What it does not do is predict the size of the effect, and it saturates before AutoAttack does.

C SUPPLEMENTARY ANALYSIS OF THE CLEAN-FEATURE ANCHOR

C.1 TEACHER EVALUATION ON PERTURBED INPUTS

The comparison in Section 3.2 evaluates the teacher at clean inputs. Here we examine why its responses to perturbed inputs may be unsuitable as supervisory targets. Under $\epsilon = 8/255$, the natural teacher’s feature rotates by 63.8° and its norm grows by a factor of 2.45. Its representation can therefore change substantially within the perturbation ball.

As a control, replacing the clean target $\Phi_t(x)$ with $\Phi_t(x')$ makes an exact copy of the teacher, $\Phi_s = \Phi_t$, a zero-loss solution even when the teacher is sensitive to perturbations. This variant yields 0.00% AutoAttack accuracy (Table 9). With a fixed clean target, copying the teacher instead incurs a loss wherever its features change within the ball.

Table 9: Where the teacher is evaluated. CIFAR-100, ResNet-18, 50 epochs; the two rows differ only in the teacher input used by the loss and inner maximization.

Teacher target	Clean	AA	NRR
$\Phi_t(x)$	61.33	26.19	36.71
$\Phi_t(x_{\text{adv}})$	76.11	0.00	0.00

IGDM (Lee et al., 2025) uses the finite difference $f_t(x + \delta) - f_t(x - \delta)$ to approximate the teacher’s response along the perturbation direction. Its accuracy depends on local linearity. The reported Taylor remainder proportion is 0.012 for adversarially trained teachers; we obtain 0.0111 for an adversarially trained ResNet-18 and 0.4763 for the natural teacher (Table 14). For the natural teacher, the finite difference has norm 14.14, compared with $\|f_t(x)\| = 18.36$, and its softmax assigns 0.917 probability to one class. These measurements indicate a less accurate local linear approximation and a highly concentrated target distribution.

We also evaluate published distillation objectives with the natural teacher replacing their robust teacher, using each method’s published recipe (Table 19). All evaluated objectives perform below adversarial training initialized from that teacher. This comparison complements the target sweep in Section 3.2, although differences among training recipes prevent it from isolating the target alone.

C.2 ADDITIONAL EVIDENCE FOR THE ANCHOR OBJECTIVE

We measure feature geometry on the final 150-epoch CFA model and a label-trained student with the same natural initialization, weight averaging, and AWP. Both models are evaluated on 5,000 test images under the same 10-step true-label PGD attack at $8/255$; all angular statistics use unit-normalized features.

Proposition 1 bounds variation within an anchored model, but does not predict a smaller variation than label training. Under the matched true-label PGD protocol, the anchor and label-trained students move by 0.312 and 0.281 of their clean feature norms. These displacements are not the

worst-case diameter O and provide no evidence of smaller relative displacement under anchoring. The reproducible difference is instead within-class concentration: CFA has a $5.3\times$ smaller clean S_w/S_b and a $2.85\times$ smaller ratio under attack (Table 10). Its class centroids are not farther apart than those learned with labels, so the result supports tighter clusters rather than greater inter-class separation.

Table 10: Feature geometry of the final models under the same true-label PGD attack. Angles are measured on unit-normalized features. *Own* is the sample-to-own-centroid angle; *gap* is the mean angle from each class centroid to its nearest other-class centroid.

	Clean			Adversarial		
	S_w/S_b	Own	Gap	S_w/S_b	Own	Gap
CFA	0.709	26.3°	29.2°	1.969	33.5°	19.2°
Label CE	3.744	52.2°	33.5°	5.611	55.0°	28.2°

The classification benefit is evaluated in Section 5.4. Without weight averaging, AWP, or the sensitivity-matched radius, the anchor improves clean and AutoAttack accuracy over label cross-entropy by 6.69 and 5.91 points under the same training conditions. The gains are 4.63 and 4.45 points when cross-entropy uses its own optimizer and schedule (Table 5, Table 25). At the full training configuration, the AutoAttack gain over cross-entropy with weight averaging and AWP is 2.40 points. Under a matched attack, CFA has slightly larger relative feature displacement and smaller centroid gaps, but substantially tighter class-conditional features. This association does not establish concentration as the cause of the robustness gain.

The checkpoint ladder in Figure 3 further separates within-class concentration from between-class separation on unit-normalized features. From the 50-epoch to the 300-epoch teacher, the mean angle from a sample to its class centroid decreases from 34.6° to 30.9° , whereas the nearest-other-centroid angle between class means increases from 35.7° to 44.9° . The corresponding students reproduce the latter change—their nearest-centroid separation increases from 25.4° to 29.8° —while their mean within-class angle remains nearly unchanged (27.9° to 28.2°). These measurements motivate interpreting the ladder through between-class geometry, but they remain observational because the teacher’s accuracy and geometry change together. The numerical values underlying the two analysis figures are collected below.

The target comparison uses the 50-epoch base regime at $\epsilon = 8/255$, without weight averaging or AWP.

Table 11: Numerical values for the target comparison in Figure 2. $H(q_t^\tau)$ is measured in nats, and NRR is the harmonic mean of clean and AA accuracy.

Target	τ	$H(q_t^\tau)$	Clean	AA	NRR
<i>Teacher-only temperature</i>					
KL	1	0.76	58.26	20.84	30.70
KL	2	2.40	59.33	22.79	32.93
KL	4	4.20	59.39	24.48	34.67
KL	8	4.56	59.47	25.19	35.39
KL	16	4.60	57.78	24.00	33.91
<i>Shared-temperature KD</i>					
KL	4	4.20	60.83	25.05	35.49
KL	8	4.56	60.99	25.25	35.71
KL	16	4.60	61.85	25.39	36.00
KL	32	4.60	61.81	25.50	36.11
Logit regression, fixed head	—	—	62.59	25.69	36.43
Feature anchor, fixed head	—	—	62.72	25.88	36.64

Table 12: Numerical values for the checkpoint trends in Figure 3.

Checkpoint	Teacher		Student		
	Clean	S_w/S_b	Clean	S_w/S_b	AA
50 epochs	75.81	1.100	64.28	0.955	24.38
100 epochs	76.62	0.982	63.65	0.900	25.20
150 epochs	77.52	0.887	62.93	0.875	25.78
200 epochs	77.66	0.808	62.72	0.850	25.88
300 epochs	78.32	0.712	62.24	0.829	25.80

C.3 SCOPE OF THE ANCHOR BOUND

Proposition 1 concerns exact maxima over the perturbation set. A loss evaluated using a finite attack does not certify these maxima. The factor 3 combines $F \leq L$ with $O \leq 2L$; the latter bound is attained when two student features lie on opposite sides of the target, each at distance L .

The argument applies to any fixed target in a normed space, including logits when matched using a norm distance. It explains how one anchor controls two quantities, but does not establish that features are superior to logits. The feature choice is motivated by Section 3.2. Likewise, the bound does not directly lower-bound robust classification accuracy: this requires information about the classifier and class margins, as discussed in Appendix B.

Joint control also does not imply that fidelity and stability can both be made arbitrarily small over the dataset. When perturbation balls overlap but their teacher targets differ, they impose conflicting constraints on the same student feature. The proposition describes how the objective controls the two errors without eliminating this possible conflict.

D ADDITIONAL RESULTS

The following tables provide additional results supporting the main experiments and supplementary analysis.

Regression under weight averaging and AWP. Table 13 compares logit and feature regression with the same natural teacher, inherited frozen classifier, AdamW optimizer, one-cycle schedule, WA, and AWP. Both use a uniform radius: neither run includes sensitivity-based allocation. Each objective is maximized to generate training perturbations and minimized to update the backbone, so the comparison includes its effect on both stages. Logit regression remains competitive with the training stack; the single-seed difference does not establish a general advantage for feature regression.

Table 13: Fixed-head regression with WA and AWP: CIFAR-100, ResNet-18, 100 epochs, uniform $\epsilon_{\text{train}} = 8.8/255$. Evaluation uses AA at $8/255$. Both losses sum squared errors across output coordinates and average over the batch.

Target	Clean	AA
Clean logits	60.36	28.02
Clean features	60.42	28.42

Table 14: Local linearity, which IGDM’s construction requires of the teacher: remainder proportion of the first-order Taylor expansion over an $\epsilon = 8/255$ ball, computed as in IGDM’s Sec. 3.1.

Network	Remainder proportion
IGDM’s robust teachers (LTD / BDM-AT / IKL-AT, published)	0.012 / 0.012 / 0.016
PGD-AT ResNet-18 (ours)	0.0111
Natural teacher (ours)	0.4763
Anchored student, $p = 0$	0.0029
Anchored student, shipped recipe	0.0202

Table 15: What to do with the classifier. CIFAR-100 base regime (50 epochs, raw features, no weight averaging, no AWP, $\epsilon = 8/255$), so that no stack component can absorb the difference.

	Clean	PGD-20	CW	AA	NRR
Logit target only, $\tau = 1$	58.26	22.62	22.79	20.84	30.70
Logit target only, $\tau = 4$	59.39	30.30	26.84	24.48	34.67
Logit target only, $\tau = 16$	57.78	31.57	26.11	24.00	33.91
Feature anchor + head KD, $\tau = 1$	62.34	26.34	26.11	23.93	34.58
Feature anchor + head KD, $\tau = 16$	62.61	32.47	27.32	25.61	36.35
Feature anchor, head untouched	62.72	28.69	27.80	25.88	36.64
head refit, adversarial CE	62.77	29.93	27.66	25.65	36.42
head refit, adv. CE + smoothing 0.1	62.57	30.43	27.55	25.66	36.39
head refit, clean CE	62.70	28.90	27.23	25.15	35.90

Normalization check. Because Section 3.4 analyzes angular geometry, we check whether discarding feature magnitude changes the training conclusion. Normalizing both sides changes NRR by 0.18 with weight averaging and AWP, while raw regression is better in the base regime. We therefore retain raw squared ℓ_2 as the default and use normalized features only for analysis; CFA does not rely on a cosine objective.

Table 16: Effect of feature normalization in the anchor loss. Normalizing both representations to the unit sphere removes their magnitudes and keeps only the angle, for which the squared Euclidean loss satisfies $\|\hat{\Phi}_s - \hat{\Phi}_t\|_2^2 = 2 - 2 \cos \angle(\Phi_s, \Phi_t)$ and is therefore equivalent to cosine loss up to an affine transformation. CIFAR-100, ResNet-18, 100 epochs; within each panel, the runs differ only in how the student and teacher features are normalized. The asymmetric row normalizes the student alone, placing the two sides at different scales, and is included only as a diagnostic.

	Clean	AA	NRR
<i>(a) With the stack: WA, AWP, $\epsilon_{\text{train}} = 8.8/255$</i>			
Raw features (squared ℓ_2)	60.21	28.14	38.35
Both normalized (cosine)	61.41	28.07	38.53
<i>(b) Base regime: no WA, no AWP, $\epsilon_{\text{train}} = 8/255$</i>			
Raw features (squared ℓ_2)	62.40	24.34	35.02
Both normalized (cosine)	61.52	22.90	33.38
Asymmetric: student normalized, teacher raw	61.28	22.98	33.43

Table 17: ϵ_{train} swept with everything else fixed. Evaluation is at $\epsilon = 8/255$ in every cell.

	Clean	PGD-20	CW	AA	NRR
<i>CIFAR-100</i>					
6/255	68.07	28.40	28.16	25.82	37.44
7/255	65.61	30.34	29.64	27.44	38.70
8/255	63.89	31.41	29.83	27.78	38.72
8.8/255 [†]	62.17	32.37	30.93	28.86	39.42
10/255 [†]	59.99	32.82	30.53	28.77	38.89
<i>CIFAR-10</i>					
6/255	89.64	49.29	50.20	47.88	62.42
7/255	88.30	51.43	52.64	50.27	64.07
8/255 [†]	87.22	52.43	53.55	51.15	64.48
8.8/255 [†]	84.96	52.98	53.94	51.74	64.31
10/255 [†]	83.29	53.25	53.88	51.79	63.87

Table 18: Effect of AWP on top of weight averaging across datasets and training radii. Every run uses the sensitivity-matched radius for 100 epochs; the AWP proxy uses $\gamma = 0.005$ after a 10-epoch warmup. Within each row, AWP is the only added component.

	WA			WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
<i>CIFAR-100</i>						
8/255	63.54	26.75	37.65	63.89	27.78	38.72
8.8/255	62.30	27.69	38.34	62.17	28.86	39.42
<i>CIFAR-10</i>						
8/255	86.67	49.20	62.77	87.22	51.15	64.48
8.8/255	85.16	50.15	63.13	84.96	51.74	64.31

Table 19: Published adversarial-distillation objectives supplied with our natural teacher. The first row of each method uses its published recipe; indented rows add our teacher initialization and then without weight averaging or AWP.

	CIFAR-100			CIFAR-10		
	Clean	AA	NRR	Clean	AA	NRR
ARD	57.61	20.24	29.96	84.58	43.75	57.67
+ teacher initialization	57.96	20.22	29.98	—	—	—
RSLAD	59.68	21.30	31.40	85.15	42.91	57.06
+ teacher initialization	59.87	20.71	30.77	—	—	—
AdaAD	59.79	23.19	33.42	85.39	46.22	59.98
+ teacher initialization	59.75	22.78	32.98	—	—	—
AdaAD + IGDM	48.25	19.48	27.75	85.01	45.50	59.27
+ teacher initialization	49.67	20.15	28.67	—	—	—
PGD-AT, teacher initialization	57.73	26.46	36.29	82.23	50.72	62.74
CFA, no WA/AWP	61.29	25.36	35.88	—	—	—

Table 20: RPAT++ given the rest of our stack. Its own recipe already carries weight averaging (EMA 0.999 from epoch 105 of 200), so the missing component is the AWP proxy, added at $\gamma = 0.005$ with warmup at 10% of the run and evaluated on its own loss; the last row also starts the run at our natural teacher. CIFAR-100, ResNet-18, our reproduction, reported at both the final and the PGD-20-selected weight-averaged checkpoint.

	Final			Best by PGD-20		
	Clean	AA	NRR	Clean	AA	NRR
RPAT++ (its own recipe)	57.05	26.61	36.30	55.93	27.37	36.74
+ AWP	57.48	28.05	37.70	57.50	27.92	37.60
+ AWP + teacher initialization	57.83	27.47	37.25	57.80	27.59	37.36

E THE ALLOCATION RULE AND ITS BUDGET SOLVE

E.1 WHERE THE RULE COMES FROM

The objective is optimized over a **pixel** ball, while what it measures is a displacement in feature space, and a fixed pixel radius moves that displacement by very different amounts across samples. The allocation follows from one elementary fact: for a differentiable loss the first-order movement available inside an ℓ_∞ ball of radius e is

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_x \mathcal{L}, \delta \rangle = e \|\nabla_x \mathcal{L}\|_1, \quad (16)$$

the dual norm of ℓ_∞ being ℓ_1 . Equalizing that movement across a batch subject to a fixed total budget $\sum_i \epsilon_i = N\epsilon$ gives $\epsilon_i g_i = \text{const}$, that is $\epsilon_i \propto 1/g_i$ with $g_i = \|\nabla_x \mathcal{L}(x_i)\|_1$. The same allocation is the max-min one: if two samples move by different amounts, shifting budget from the sample that moves more to the sample that moves less raises the minimum, so no unequal assignment is optimal. Taking $\epsilon_i \propto (\bar{g}/g_i)^p$ makes $p = 1$ that allocation, $p = 0$ the uniform-pixel one, and $p \in (0, 1)$ an interpolation.

Two approximations are worth declaring rather than leaving to be found. First, g_i is measured once, by a single backward pass at the *clean* x_i before the attack starts, so it is the sensitivity at the starting point and not along the PGD trajectory. Second, everything above is first order in ϵ . Neither is repaired anywhere in the method; the rule is a budget heuristic with a derivation, not an exactly solved subproblem.

We also do not claim the rule is specific to a directional objective. $\|\nabla_x \mathcal{L}\|_1$ is defined for any loss and the implementation reads whichever loss is configured; on the shipped raw- ℓ_2 anchor it equalizes a feature displacement rather than an angle, which is why the honest name is *sensitivity-matched* ϵ rather than an angular budget. What separates it from IAAT, MMA and CAT is not directionality but the signal: they allocate from a property of the sample, this allocates from the geometry the training loss is written in.

The separation is not one of dispersion, which is worth stating because a rule that simply spread the budget more aggressively would be the simpler explanation of any gain. Measured over training on the shipped configuration, the coefficient of variation of our sensitivity signal has median 0.56, against 0.75 for the difficulty score IAAT and CAT allocate from and 1.33 for the logit margin of MMA. The box of this section compresses all three before they reach the attack, leaving per-sample multipliers of spread 0.32, 0.39 and 0.42 respectively, and the ordering survives the compression. Ours is the least dispersed of the three signals either way, so the comparison in Table 7 is a comparison of which geometry the budget follows.

One discrepancy is ours to declare rather than the reader’s to find. Equation (16) gives the ℓ_1 norm and the implementation reads ℓ_2 , for no better reason than that the cell which fixed the recipe used it and every logged run since had to stay reproducible against that choice. The two are not the same allocation. Measured over training on the shipped configuration, the ℓ_1 signal’s coefficient of variation has median 0.68 against the ℓ_2 signal’s 0.56, and the per-sample multiplier the attack receives has spread 0.34 against 0.32 — so the norms really do distribute the budget differently.

They nonetheless land close together. Training the shipped recipe with `featdir_angepts_gnorm` set to 1 and nothing else changed gives 62.16/31.97/30.56/28.57 (NRR 39.15) against 62.17/32.37/30.93/28.86 (NRR 39.42): clean accuracy is identical to a hundredth and AutoAttack differs by 0.29, in favour of the norm we ship.

We do not call that a tie. It is twice the seed-to-seed spread of Appendix F, and it is the same order of magnitude as what the allocation buys in the first place: against the uniform radius, $p = 1$ gains +1.75 clean and +0.44 AutoAttack at ℓ_2 and +1.74 and +0.15 at ℓ_1 . So the rule’s benefit survives the norm in direction and almost entirely on the clean axis, while most of its AutoAttack half does not. The honest statement is that the norm is a real choice which we made before we could compare, that the comparison favours what we ship, and that a reader preferring the derived ℓ_1 would still get the clean gain and a third of the robust one.

The permuted row of Table 7 deserves its construction spelled out, since it is what makes the comparison a comparison of signals rather than of budgets. It takes the multiset of weights the gradient rule produces — the same values, the same clip $[0.5, 1.5]$, the same restored mean — and only permutes which example receives which, so that the larger radii go to the easier examples, which is the direction IAAT and CAT assign them. Reordering costs 0.80 AutoAttack against the rule it reorders and lands below even the uniform row on AutoAttack, CW and NRR alike.

E.2 CLIPPING AND RESTORING THE MEAN RADIUS

The reported configuration floors the gradient norm at 10^{-12} and computes $v_i = \text{clip}((\bar{g}/g_i)^p, 0.5, 1.5)$, followed by $w_i = Nv_i / \sum_j v_j$ and $\epsilon_i = \epsilon w_i$. PGD uses step size $a_i = \epsilon w_i$. This preserves the batch mean radius exactly. The clipping thresholds apply before normalization, so they are not hard bounds on the final multipliers. When all gradients vanish, the floor makes all g_i equal and the allocation uniform.

Optional exact-box variant. The implementation also supports an alternative that constrains the final multipliers. It is not enabled in the reported default configuration.

With $r_i = (\bar{g}/g_i)^p$ and box $[w_{\text{lo}}, w_{\text{hi}}] = [0.5, 1.5]$, the radii that satisfy both the box and the budget exactly are $\epsilon_i = \epsilon \cdot \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}})$ with t the solution of

$$\sum_{i=1}^N \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}}) = N. \quad (17)$$

The left side is continuous and nondecreasing in t from Nw_{lo} to Nw_{hi} , so a root exists and bisection finds it; when no clip is active it reduces to $t = N/\sum_i r_i$, that is, to restoring the mean. Clipping first and rescaling afterwards, the obvious alternative, does not preserve the box—the rescaling moves the clipped entries back out of it.

Table 21: WideResNet-34-10 on CIFAR-100 at $\epsilon = 8/255$, measured in our framework.

Method	T.	Clean	AA	Avg	NRR
PGD-AT	–	58.97	22.84	40.91	32.93
TRADES	–	57.35	25.60	41.48	35.40
MART	–	57.16	22.43	39.80	32.22
ARD	nat.	59.52	23.05	41.29	33.25
RSLAD	nat.	60.48	21.63	41.06	31.87
AdaAD	nat.	61.66	25.62	43.64	36.20
LBGAT	nat.	59.63	28.55	44.09	38.61
ARREST	nat.	67.69	24.69	46.19	36.18
HAT	nat.	62.81	26.91	44.86	37.68
ADR	self	64.50	29.04	46.77	40.05
CFA (ours)	nat.	65.04	30.14	47.59	41.19

F EXPERIMENTAL DETAILS

Table 22: Natural teacher checkpoints used by the shipped models. The teachers are trained only on clean examples and have zero adversarial accuracy.

Dataset	Architecture	Epochs	Clean
CIFAR-10	ResNet-18	200	95.32
CIFAR-100	ResNet-18	200	77.66
Tiny-ImageNet	ResNet-18	200	66.29

Every number we report is the **final** epoch, never a checkpoint selected on a robustness metric. This matters when reading Table 29: RPAT selects on PGD-20 and ADR on PGD-10, so those rows are upper envelopes of a training run where ours is its endpoint.

Teacher. One naturally trained network per dataset, no adversarial examples anywhere in its training, and the student is warm-started from the same checkpoint. ResNet-18, SGD at learning rate 0.1 under a one-cycle schedule, batch size 128, no augmentation beyond the standard crop and flip, for 200 epochs on each dataset. The resulting teachers are summarized in Table 22. The teacher’s own robustness is 0 — it is never attacked and is not meant to survive attack; Figure 3 is the ablation over how long it is trained.

Student. The main ResNet-18 configurations are listed below. Unless noted otherwise, CIFAR-100 ablations use 100 epochs at $8.8/255$, with a 10-epoch AWP warmup.

	CIFAR-10	CIFAR-100	Tiny-ImageNet-200
Backbone	ResNet-18	ResNet-18	ResNet-18
Optimizer	AdamW	AdamW	AdamW
Learning rate / schedule	0.021, one-cycle	0.021, one-cycle	0.021, one-cycle
Epochs	150	150	100
Batch size	128	128	128
Training radius ϵ_{train}	8/255	8/255	8.8/255
Inner attack	PGD-10, step 2/255	PGD-10, step 2/255	PGD-10, step 2/255
Allocation exponent p	1	1	1
Radius box $[w_{\text{lo}}, w_{\text{hi}}]$	[0.5, 1.5]	[0.5, 1.5]	[0.5, 1.5]
Classifier head	frozen	frozen	frozen
Weight averaging (start, decay)	0.2, 0.999	0.2, 0.999	0.2, 0.999
AWP (γ , warmup)	0.005, 15 ep	0.005, 15 ep	0.005, 10 ep
Teacher checkpoint	clean, 200 ep	clean, 200 ep	clean, 200 ep

The head being frozen is what removes τ and β from the method: with the classifier inherited and never updated, the head-KD term of earlier drafts has no parameters to reach and is switched off entirely. p is not tuned. The training-radius sweep is reported in Table 17; evaluation is at 8/255 in every cell.

Evaluation. We report AutoAttack accuracy at ℓ_∞ radius 8/255 on the full test set; NRR is its harmonic mean with clean accuracy. Additional attacks are in Appendices I and K. Ported baselines retain their published recipes unless stated otherwise.

Repeated training. We repeat both arms of the final radius-allocation comparison with three student seeds. Uniform radii give 62.63 ± 0.22 clean and 28.01 ± 0.20 AA; sensitivity matching gives 64.82 ± 0.07 clean and 28.04 ± 0.09 AA (mean $\pm$ sample standard deviation). The paired clean gains are 2.35, 2.24, and 1.97 points, whereas the paired AA changes are 0.14, -0.23 , and 0.18. Thus the clean gain repeats, while the data support no AA change. Other rows remain individual runs, so small differences between regression targets or nearby operating points should not be interpreted as significant.

Compute, teacher included. A single RTX 5080. One student cell is about 2.2 hours on CIFAR-100 at 100 epochs, of which the AutoAttack evaluation is roughly 7 minutes; Tiny-ImageNet is longer in proportion to its test set.

The teacher has to be counted, since calling it free would otherwise hide a training run. It is not free, but it is cheap: a clean epoch has no inner maximization, so it costs 6.8 seconds against 74.5 for an adversarial one, and 200 clean epochs come to 0.39 hours. The teacher is therefore 19% of one student run and is amortized over every cell that reads it, the teacher ladder of Figure 3 included. Table 23 puts the total against the baselines measured on the same machine.

Table 23: Wall-clock training cost on one RTX 5080, CIFAR-100, ResNet-18, as logged by the runs reported in this paper. Evaluation is excluded. Adversarial distillation additionally requires an adversarially trained teacher, which is not counted in its row and is the more expensive artifact.

Method	Epochs	Teacher (h)	Student (h)	Total (h)
Natural teacher (ours, reused)	200 clean	–	0.39	0.39
CFA (ours)	100 adv.	0.39	2.07	2.46
Label CE at our regime	100 adv.	0.39	3.22	3.61
PGD-AT, initialized at the teacher	100 adv.	0.39	1.72	2.11
ARD	100 adv.	0.39	1.52	1.91
RSLAD	100 adv.	0.39	1.71	2.10
AdaAD	100 adv.	0.39	2.79	3.18
LBGAT	100 adv.	joint	1.88	1.88
HAT	50 adv.	0.39	1.10	1.49
ADR	200 adv.	self	3.52	3.52

Two readings follow. The anchor is not bought with compute, since it costs less end to end than label cross-entropy at the same regime and about what PGD-AT costs once the teacher is included. And

the comparison with adversarial distillation is favourable in a way the table understates, because those rows assume a robust teacher that the current literature obtains from a WideResNet trained on generated data, which is more expensive than every row here combined.

G ADDITIONAL TEACHER-CHECKPOINT RESULTS

Figure 3 shows one natural-training trajectory; Table 24 reports an independently trained trajectory. Each listed teacher was trained to completion under its own one-cycle schedule, then used to initialize a 50-epoch student with the same training procedure. These experiments vary the teacher checkpoint, not just the target features.

Table 24: Independent CIFAR-100 teacher trajectory (teacher seed 1; 50-epoch students with seed 0). Teacher metrics are measured on held-out data.

Epoch	Teacher			Student		
	Clean	Margin	Sensitivity	Clean	AA	NRR
20	73.87	3.074	3.911	63.82	23.22	34.05
40	75.57	3.931	4.542	64.36	23.86	34.81
100	77.27	4.506	4.678	63.68	25.61	36.53
150	77.68	4.495	4.262	63.37	25.49	36.36
200	77.90	4.542	4.129	63.02	26.16	36.97
300	78.27	4.523	3.835	62.34	25.94	36.64

The second trajectory again moves from higher student clean accuracy at earlier checkpoints to higher AA at later ones. The strongest AA occurs at 200 rather than 300 epochs, although the difference is small. Feature sensitivity has only moderate rank correlation with student clean accuracy on these six checkpoints (Spearman 0.54, versus 0.83 on the same grid of the first trajectory). The matched teachers in Table 2 further show that teacher clean accuracy alone does not determine student clean accuracy. None of these measurements supplies a validated teacher-only selection rule.

H THE RECIPE CONTROLS

Unless explicitly ablated, CFA warm-starts from the natural teacher and trains with AdamW at 0.021 under a one-cycle schedule, while the baselines retain their authors’ recipes. Table 25 asks whether that recipe is what wins: it runs our own objective, and label cross-entropy for reference, under both recipes and both initializations. Every row is measured here at the base regime, which carries no weight averaging, no AWP and a uniform radius, on CIFAR-100 with ResNet-18 for 100 epochs.

Table 25: The recipe controls. *Ours* is AdamW at 0.021 under a one-cycle schedule with the teacher warm start; *theirs* is SGD at 0.1 decayed by ten at epochs 70 and 90, the schedule PGD-AT uses here. The training radius and every other field are held fixed within each pair.

Objective	Recipe	Clean	AA	Avg	NRR
Anchor	ours, teacher init	61.29	25.36	43.33	35.88
Anchor	ours, random init	60.81	25.21	43.01	35.66
Anchor	theirs, teacher init	59.65	25.54	42.60	35.77
Label cross-entropy	ours, teacher init	54.60	19.45	37.03	28.68
Label cross-entropy	theirs, teacher init	56.66	20.91	38.79	30.55

Changing initialization or schedule changes the anchor’s NRR by 0.22 and 0.11, respectively. Cross-entropy improves by 1.87 NRR under its own schedule, so Table 5 includes that stronger baseline. The anchor remains ahead under both tested schedules. These comparisons support an objective-level contribution within the tested recipes, without establishing significance from single runs or claiming exhaustive tuning of the alternatives.

Table 26: Successive additions to the CIFAR-100 ResNet-18 anchor (100 epochs, $\epsilon_{\text{train}} = 8.8/255$).

	Clean	AA	NRR
Anchor ($p = 0$)	61.29	25.36	35.88
+ sensitivity-matched ϵ	63.09	25.74	36.56
+ weight averaging	62.30	27.69	38.34
+ AWP	62.17	28.86	39.42

I GRADIENT MASKING

We test for gradient masking with AutoAttack, independent attacks, and PGD sweeps (Athalye et al., 2018). Table 27 reports these checks for a 100-epoch CIFAR-100 repeat with 62.00% clean accuracy.

Table 27: Gradient-masking checks on all 10,000 CIFAR-100 test images at 8/255. Individual attacks are evaluated independently.

Check	Setting	Accuracy
<i>white box against black box</i>		
APGD-CE	white box, untargeted	31.80
APGD-T	white box, targeted	28.83
FAB-T	white box, boundary	29.16
Square	black box , 5000 queries	34.82
AutoAttack	standard evaluation	28.73
<i>radius sweep, PGD-20</i>		
$\epsilon = 2/255$		54.62
$\epsilon = 4/255$		47.03
$\epsilon = 8/255$		32.46
$\epsilon = 12/255$		20.88
$\epsilon = 16/255$		11.94
$\epsilon = 32/255$		1.05
$\epsilon = 64/255$		0.00
<i>step sweep, $\epsilon = 8/255$</i>		
PGD-10 / 20 / 50 / 100		33.17 / 32.57 / 32.30 / 32.24
<i>transfer</i>		
Natural teacher $\rightarrow$ student	PGD-20 built on the teacher	60.42

AutoAttack yields 28.73% accuracy. PGD accuracy falls monotonically with attack radius, reaching zero at 64/255, and decreases with additional steps. These checks show no indication of gradient masking under the evaluated attacks.

The transfer row is worth reading separately. The student inherits the teacher’s classifier and is initialized at the teacher’s features, so adversarial examples built on the teacher might be expected to carry over; they cost the student 1.58 points against its clean accuracy. Sharing a head is therefore not sharing a vulnerability, which is the same conclusion Section 5.4 reaches from class separation: what the anchored student changes is the position of its features, and the teacher’s own adversarial directions do not point at the student’s boundaries.

J NOTES ON THE BASELINES

Several rows of Table 3 run a specific variant of the method they name, or are grouped in a way that needs saying. This section is those decisions; the table carries none of them.

ARREST uses its published objective: adversarial finetuning from the natural model with SGD at 0.025, the angular representation term at $\lambda = 50$, and Noisy Replay over the first half at $\phi = 30^\circ$. Table 3 reports its 150-epoch run for the main comparison, whereas Table 8 uses its published 20-epoch schedule. We read its representation after global pooling, where our anchor and geometry measurements are defined.

LBGAT, and why its CIFAR-10 cell is empty. The authors release a training script for CIFAR-100 only, and their CIFAR-10 entries are released as pretrained WideResNet checkpoints. Running their CIFAR-100 recipe on CIFAR-10 diverges here within thirty steps: the squared error is reduced with mean over batch and classes, so a ten-class problem receives ten times the gradient of a hundred-class one at equal logit error, and $\text{lr} = 0.1$ is above the stability threshold that follows. Lowering the learning rate trains (at 0.01 the natural branch reaches 49% within 180 steps), but a learning rate we chose is not their recipe, so the cell is left empty rather than filled with a number of our own construction. LBGAT trains its natural branch jointly with the robust one, so unlike the distillation rows its teacher is not obtained for free, and we group it accordingly. The two datasets run different variants because the authors publish different ones: CIFAR-100 is LBGAT6 and CIFAR-10 is LBGAT0, their release lists CIFAR-10 checkpoints under LBGAT0 alone, and $\beta = 6$ is their CIFAR-100 setting. We report the configuration each dataset has a published counterpart for rather than forcing one β across both.

HAT uses a naturally trained network only to *label* helper points beyond the ball, never as a target, so it belongs with the trade-off methods and not the distillation ones. It is run on its own one-cycle schedule rather than ours, which collapses it to chance on both datasets — the same allowance we make for LBGAT. On CIFAR-10 the port reproduces HAT’s published no-extra-data figure to within 0.31 clean and 0.27 AutoAttack (84.90/49.08 reported, ResNet-18), which is what licenses reporting its CIFAR-100 row at all: HAT publishes no CIFAR-100 result without additional data, its 61.50/28.88 being trained with DDPM-generated samples, so that column has no published figure to be read against.

ADR is run on separate hardware and its row is filled from there. Two things have to hold when it is. ADR reports two configurations — AT+ADR at 56.10/26.87 on CIFAR-100 and AT+WA+AWP+ADR at 57.36/28.50 — and the row has to name which, the second being the one component-matched to our recipe, which carries both weight averaging and AWP. ADR is also self-distillation: its teacher is an exponential moving average of the student, no external network is read, and the student trains from scratch, so its row inherits nothing of our natural teacher.

What the published tables are blocked by. Table 29 and Table 30 separate their rows by network, and the separations are not cosmetic. RPAT’s “ResNet-18” is `ResNet(Bottleneck, [2, 2, 2, 2])` in its released code, 14.35M parameters against the 11.27M of the standard `BasicBlock` network used by us and by ADR; they also ship a `pre_resnet18` at 11.27M and use `PreActResNet-18` explicitly elsewhere, so the distinction is theirs. HAT’s Tiny-ImageNet row is `PreActResNet-18`: ADR lists it under ResNet-18, but the figures are quoted from HAT’s own Table 9, whose caption says `PreAct`, and HAT’s code refuses anything else on that dataset — same parameter count as ours, different network, and its two companions from that table are shown beside it. F²AT’s plain SAT baseline reaches 31.52/9.37 where ADR’s plain AT reaches 45.87/18.06, a gap no difference of method explains, so its rows are separated too. TE’s architecture we could not verify — it is ADR’s attribution to a paper we do not have — and it is marked accordingly. ADR trains Tiny-ImageNet for 80 epochs against our 100, which favours us and is stated here rather than left implicit. LBGAT reports that dataset without AutoAttack and B-MTARD reports it on MobileNet-v2, so neither appears; CURE, ARREST and Generalist++ do not report it at all.

IGDM is reported at $\alpha = 1$. Its published $\alpha = 20$ diverges to chance accuracy on both datasets when the teacher is natural, as documented in Table 19.

Teacher-only KL with the training stack. This diagnostic combines probability-target distillation with WA and AWP; it is not a reproduction of a published method. Temperature is applied only to the teacher, and the best tested stacked setting, $\tau = 4$, gives 56.97/26.96 clean/AA; $\tau = 16$ gives 42.45/20.10. The comparison concerns this KL training design, not all logit-based supervision. In particular, fixed-head logit MSE is evaluated separately in Table 13, where it remains competitive with feature regression.

K PER-ATTACK METRICS

Table 28 gives every attack we measure for our own cells. The main tables report clean accuracy and AutoAttack because those are the two axes the trade-off is defined on, and because PGD-20 is not a faithful reading of this method.

Table 28: Per-attack accuracy (%) for our cells. ResNet-18, $\epsilon = 8/255$, final-epoch weight-averaged model, full test set.

Dataset	Cell	Clean	FGSM	PGD-20	CW	AA	NRR
CIFAR-100	CFA, head frozen (shipped)	62.17	36.91	32.37	30.93	28.86	39.42
	the same cell at seed 1	62.00	36.79	32.49	30.82	28.74	39.27
	ℓ_1 gradient norm	62.16	36.48	31.97	30.56	28.57	39.15
	CFA, head-KD term retained	62.35	38.80	36.26	30.65	28.68	39.29
	CFA, $\epsilon_{tr} = 6/255$	68.07	35.88	28.40	28.16	25.82	37.44
	CFA, $\epsilon_{tr} = 7/255$	65.61	36.32	30.34	29.64	27.44	38.70
	CFA, $\epsilon_{tr} = 8/255$	63.89	36.67	31.41	29.83	27.78	38.72
	CFA, $\epsilon_{tr} = 10/255$	59.99	36.50	32.82	30.53	28.77	38.89
	CFA, no WA / AWP	61.29	33.97	28.23	27.19	25.36	35.88
	Teacher-only KL, our stack	56.97	35.82	33.68	28.94	26.96	36.60
	RPAT++ (our reproduction)	55.93	34.46	32.44	29.37	27.36	36.74
	RPAT++ (our reproduction)	55.93	34.46	32.44	29.37	27.36	36.74
CIFAR-10	CFA, $\epsilon_{tr} = 8.8/255$ (shipped)	84.96	59.58	52.98	53.94	51.74	64.31
	CFA, $\epsilon_{tr} = 6/255$	89.64	59.37	49.29	50.20	47.88	62.42
	CFA, $\epsilon_{tr} = 7/255$	88.30	59.59	51.43	52.64	50.27	64.07
	CFA, $\epsilon_{tr} = 8/255$	87.22	59.92	52.43	53.55	51.15	64.48
	CFA, $\epsilon_{tr} = 10/255$	83.29	58.98	53.25	53.88	51.79	63.87
	RPAT++ (our reproduction)	82.41	59.73	55.72	52.88	50.75	62.82

Why PGD-20 is not the metric here. The student inherits the teacher’s classifier and never retrains it. PGD-20 ascends the cross-entropy *through that classifier*, so it reads a head that was fitted for a different feature distribution; AutoAttack, which includes a targeted APGD, and CW_∞ , which attacks the logit margin directly, do not depend on that fit in the same way. The consequence is measurable on a controlled pair. The first two rows of Table 28 share a backbone objective, a schedule and an attack, and differ only in whether the classifier is refit:

	PGD-20	CW	AA
head-KD term retained	36.26	30.65	28.68
head frozen	32.63	30.66	28.77
difference	−3.63	+0.01	+0.09

Changing the classifier moves PGD-20 by 3.63 points and moves CW and AutoAttack by hundredths. The same pattern recurs across our ablations—the head-temperature sweep, the label-smoothing sweep, the directional-versus-raw comparison, and the radius sweep all show PGD-20 moving against CW and AutoAttack. We therefore report PGD-20 for completeness and read the trade-off on clean accuracy and AutoAttack, which is also what ADR and ARREST report in their main tables.

L PUBLISHED RESULTS, FOR CONTEXT

Table 3 contains only numbers we measured ourselves. Table 29 collects the published figures for the same and neighbouring methods, so that our runs can be placed against the literature without the two being mixed in one column. These are quoted as reported and were produced by other codebases, other schedules and other checkpoint-selection rules—RPAT selects its best checkpoint by PGD-20 and ADR by PGD-10, where every number of ours is the final epoch—so they should be compared within a source rather than across.

Read against Table 3 and restricted to the ResNet-18 blocks, which are the ones our numbers are comparable with, our CFA row exceeds every published Avg and NRR on both datasets: 64.82/28.04 at Avg 46.43 on CIFAR-100 and 87.37/51.33 at Avg 69.35 on CIFAR-10.

Table 29: Published results, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. **The architecture changes between blocks** and is named in each heading; only the ResNet-18 blocks are comparable with Table 3. These were produced by other codebases, schedules and checkpoint-selection rules, so they should be compared within a source rather than across.

Method	Teacher	Clean	AA	Avg	NRR	Source
<i>CIFAR-100, ResNet-18</i>						
PGD-AT	–	56.56	25.02	40.79	34.69	RPAT
TRADES	–	55.39	24.51	39.95	33.98	RPAT
MART	–	49.83	25.00	37.41	33.30	RPAT
Consistency-AT	–	58.53	25.39	41.96	35.42	RPAT
Consistency-AT + RPAT	–	60.33	26.31	43.32	36.64	RPAT
F ² AT	–	54.19	23.24	38.71	32.53	F ² AT
Generalist++	co-tr.	62.97	23.96	43.47	34.71	Generalist++
ADR (AT + ADR)	self	56.10	26.87	41.48	36.34	ADR
ADR (AT + WA + AWP + ADR)	self	57.36	28.50	42.93	38.08	ADR
<i>CIFAR-10, ResNet-18</i>						
PGD-AT	–	82.78	44.63	63.71	57.99	CURE
TRADES	–	82.41	48.37	65.39	60.96	CURE
MART	–	80.70	47.49	64.10	59.79	CURE
ST-AT	–	83.10	50.50	66.80	62.82	CURE
IAD	robust	80.63	50.17	65.40	61.85	CURE
Consistency-AT + RPAT	–	84.12	48.98	66.55	61.91	RPAT
Generalist++	co-tr.	89.09	46.07	67.58	60.73	Generalist++
ARREST	natural	86.63	46.14	66.39	60.21	ARREST
CURE	self	86.76	49.69	68.23	63.19	CURE
ADR (AT + ADR)	self	82.41	50.38	66.40	62.53	ADR
<i>CIFAR-10, PreActResNet-18</i>						
WA (Izmailov et al., 2018)	–	83.50	49.89	66.70	62.46	RPAT
AWP (Wu et al., 2020)	–	81.11	50.09	65.60	61.93	RPAT
KD + SWA (Chen et al., 2021)	robust	84.06	49.82	66.94	62.56	RPAT
TE (Dong et al., 2022)	–	82.04	50.12	66.08	62.23	RPAT
ReBAT (Wang et al., 2023)	–	82.09	50.72	66.41	62.70	RPAT
RPAT++ (Wang & Liu, 2025)	–	82.63	51.00	66.82	63.07	RPAT
<i>CIFAR-100, PreActResNet-18</i>						
WA (Izmailov et al., 2018)	–	57.26	25.83	41.55	35.60	RPAT
AWP (Wu et al., 2020)	–	54.10	25.16	39.63	34.35	RPAT
KD + SWA (Chen et al., 2021)	robust	57.17	25.66	41.42	35.42	RPAT
TE (Dong et al., 2022)	–	56.41	25.84	41.13	35.44	RPAT
ReBAT (Wang et al., 2023)	–	56.13	27.60	41.87	37.00	RPAT
RPAT++ (Wang & Liu, 2025)	–	56.84	27.68	42.26	37.23	RPAT
<i>CIFAR-10, WideResNet-34-10</i>						
WA (Izmailov et al., 2018)	–	87.66	52.65	70.16	65.79	RPAT
MMA (Ding et al., 2020)	–	87.80	43.10	65.45	57.82	RPAT
AWP (Wu et al., 2020)	–	85.63	53.32	69.48	65.72	RPAT
KD + SWA (Chen et al., 2021)	robust	87.45	53.59	70.52	66.46	RPAT
TE (Dong et al., 2022)	–	85.97	52.88	69.43	65.48	RPAT
ReBAT (Wang et al., 2023)	–	85.25	54.78	70.02	66.70	RPAT
ADR (Wu et al., 2024)	self	84.67	53.25	68.96	65.38	RPAT
CURE (Gowda et al., 2024)	self	87.05	52.10	69.58	65.19	RPAT
RPAT++ (Wang & Liu, 2025)	–	86.76	54.97	70.87	67.30	RPAT

The restriction is not a convenience. The WideResNet-34-10 block exceeds ours on both criteria—RPAT++ reaches Avg 70.87 and NRR 67.30 there—because a 48.3M network is a different model, not a better method, and Tables 4 and 21 are where that comparison belongs. The same caution applies to the PreActResNet-18 blocks in the other direction: RPAT++ falls to 82.63/51.00 there, and reading that as a defeat for RPAT would be the identical error. We state all of this as a comparison across codebases and architectures rather than as a controlled result, which is what Table 3 is for.

Table 30: Published Tiny-ImageNet-200 results, ResNet-18, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. Our row is above all of them on both axes. The blocks are separated because the networks are not the same, and Appendix J says how each differs.

Method	Clean	AA	Avg	NRR	Source
CFA (ours)	55.16	20.54	37.85	29.93	—
AT + WA + ADR	48.55	20.23	34.39	28.56	ADR
AT + WA + AWP + ADR	48.27	20.12	34.20	28.40	ADR
TRADES + WA + AWP + ADR	51.38	19.48	35.43	28.25	ADR
TRADES + WA + ADR	51.99	19.17	35.58	28.01	ADR
AT + WA + AWP	48.61	19.58	34.09	27.92	ADR
AT + WA	49.10	19.30	34.20	27.71	ADR
TE*	47.46	18.29	32.88	26.40	ADR
AT	45.87	18.06	31.96	25.92	ADR
TRADES	48.49	17.35	32.92	25.56	ADR
<i>PreActResNet-18 — same parameter count, different network</i>					
HAT	52.60	18.14	35.37	26.98	HAT
TRADES	48.25	17.17	32.71	25.35	HAT
AT	47.76	17.92	32.84	26.02	HAT
<i>Bottleneck ResNet-18 (14.35M) and an unstated setup — see caption</i>					
Consistency-AT + RPAT	49.74	18.84	34.29	27.33	RPAT
PGD-AT + RPAT	47.68	17.77	32.73	25.89	RPAT
Consistency-AT	46.54	17.60	32.07	25.54	RPAT
MART	39.70	17.18	28.44	23.98	RPAT
F ² AT	40.54	13.13	26.84	19.84	F ² AT
SEAT	35.51	11.37	23.44	17.22	F ² AT
SAT	31.52	9.37	20.45	14.45	F ² AT

M THE CURE REPRODUCTION

Seven runs of the official CURE repository, from its own code and our shared CIFAR data, none of which reproduce the published 86.76/49.69 on CIFAR-10. They fall short on *different* axes: the closest on standard accuracy reaches 86.11 at AA 40.65, the closest on robustness AA 45.63 at clean 81.15, while the published pair has both. Running the repository unmodified gives 65.03/29.74. Computing its RGP prominence mask on absolute values raises AA by 15.9 points (29.74 $\rightarrow$ 45.63), which localizes the discrepancy to that step without closing it.

We re-evaluated all five surviving checkpoints with our own pipeline and it agrees with the repository’s own evaluation to within 0.05 AutoAttack on every one, so the gap is in training rather than in measurement. For contrast, RPAT++ reproduces to within 0.31 AutoAttack of its published numbers on both datasets under the same conditions. The CURE row of Table 29 is therefore the paper’s own number, and CURE is omitted from Table 3.

N A CORRECTION TO THE AWP PROXY, AND WHAT IT MOVED

AWP ascends an auxiliary objective in weight space before each descent step. In our implementation the closure used for that ascent had drifted from the closure used for descent, in two ways. It retained a head-distillation term that `freeze_head` removes from training, so AWP was maximizing a quantity partly composed of a term the model never optimizes; and a detach intended for the normalized feature was applied on a path that feeds the unnormalized one. Measured on CIFAR-100 before the fix, the spurious term was 7.15 against the anchor’s 4.15 and contributed 40.8% of the

AWP backbone gradient, at cosine 0.918 to the correct direction. Separately, the head was frozen after the proxy was copied, which is harmless only because the proxy’s first step is delayed by the AWP warmup.

We report this because it is the kind of defect that changes a paper’s numbers, and because here it did not. Re-running the shipped recipe with both closures agreeing gives 62.17/28.86 against 62.65/28.77 on CIFAR-100 and 84.96/51.74 against 85.58/51.79 on CIFAR-10: NRR moves by 0.01 and 0.22, no ordering in any table changes, and clean accuracy moves more than robustness does. AWP is evidently insensitive to a large, well-aligned perturbation of its ascent direction, which is consistent with its role here as a regularizer of sharpness rather than a precisely targeted objective. All numbers reported for the shipped recipe are post-correction. Two exceptions are stated where they appear: the head-freeze comparison in Appendix K and the ϵ_{train} sweep in Table 17 were measured before it. In each, every arm shares the defect, so the comparison is internally consistent even though its absolute values differ from the corrected ones by a few hundredths.

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